



ScienceDirect

Journal of Hydrology

Date: December 2025

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A multivariate copula framework quantifies the augmenting joint risks of compound high temperature and dry/wet events under climate change in China

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Metrics

Highlights

- In the future, CHTDE and CHTWE are expected to increase by an average of 25–55% and 92–677%, respectively.
- Approximately 87% and 93% of regions will experience more frequent, larger, and intense of CHTDE and CHTWE.
- The three-dimensional Copula results more prominently highlight the severity of compound events compared to the two-dimensional Copula.

Outline

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Abstract

As climate change intensifies, the urgency to accurately quantify the risks associated with compound climate events increases, yet the multi-dimensional integrated risk assessment of these events remains very weak. Here, we assessed the occurrence characteristics of compound high temperature and dry events (CHTDE) and compound high temperature and wet events (CHTWE) in the historical and future using daily data from the Coupled Model Intercomparison Project Phase 6 (CMIP6), and constructed a three-dimensional copula model to investigate their joint occurrence risks in China. In future emission scenarios, the precipitation threshold line are projected to shift northward, and the extent of high temperatures will continuously expand. These changes directly influenced the spatial distribution of two types of compound events (CEs). Under the highest emission scenario, China is likely to experiences an increase in the total/maximum duration, frequency, and magnitude of both types of CEs, with CHTDE and CHTWE rising by 25%-55% and 92%-677%, respectively. If emissions were controlled under the SSP245 scenario, the occurrence of CHTDE could decrease overall by 11.5%-30%. Specifically, 87% and 93% of regions will experience more frequent, larger, and more intense of CHTDE and CHTWE. Historically, CEs were concentrated in July and August, but it is likely to occur more frequently across a broader range of months in future. The three-dimensional copula analysis highlights the complex interdependencies among the duration, frequency, and intensity of CEs under different climate scenarios. Our findings further emphasize the increase in severity of CEs in future scenarios, with a noticeable reduction in their return periods, compared to two-dimensional copula analysis. We quantitatively assessed the occurrence risk of CEs, explored the mechanisms of their occurrence and persistence, and emphasized that CHTWE should warrant greater attention.

Keywords

Compound events; CMIP6; Copula; China

5. Conclusion

7. Consent for publication

Author contributions

9. Plain Language Summary

CRedit authorship contribution statement

Ethics approval and consent to participate

Declaration of competing interest

Acknowledgments

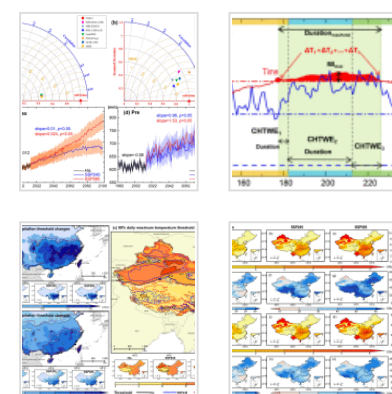
Appendix A. Supplementary data

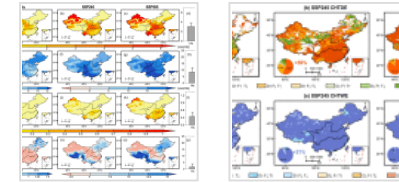
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1. Introduction

Amid global warming, the concern over extreme weather events has grown significantly. In particular, compound events (CEs)—characterized by the simultaneous or sequential interaction of multiple extreme climate hazards—are expected to exacerbate under climate change (AghaKouchak et al., 2014, Zscheischler et al., 2018). Such CEs have been shown to cause far more destructive impacts on human society and ecosystems than single extreme events (Fischer et al., 2021, Zscheischler et al., 2020).

The combination of high temperature and precipitation has long been a focal point in the study of CEs. In recent years, research on compound high temperature and dry events (CHTDE) has proliferated, demonstrating that these events significantly trigger forest wildfires, vegetation mortality, and a range of disruptions to production and daily life (Bevacqua et al., 2022, Geirinhas et al., 2021, Li et al., 2021a, Li et al., 2021b, Liu et al., 2022, Wu et al., 2020). Compound high temperature and wet events (CHTWE) represent another type of CEs associated with extreme high temperature, which conditions significantly impair human thermoregulation via evaporative cooling, leading to elevated perceived temperatures. These events typically cause substantial population exposure, exacerbate risks of crop diseases and pests, and may exert far greater potential impacts on vegetation and ecosystems than individual extreme temperature or precipitation alone (Ahmed and Wang, 2021, Coffel et al., 2017, Saeed et al., 2021, Zhang et al., 2021a). Since the onset of the 21st century, the probability of such events has increased by a factor of 5 to 10 (Chen et al., 2021). In the past five years, occurrences of CHTWE have been documented across China (Chen et al., 2021), Western Europe (Cornwall, 2021), and Pakistan (Mallapaty, 2022, Nanditha et al., 2023). However, current academic research has paid relatively little attention to the assessment of CHTWE based on reliable observations and numerical models. Furthermore, limiting the study object to individual CEs may constrain systematic evaluation of compound extremes. Therefore, conducting

Extras (2)

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comparative studies on these two classes of CEs—both related to extreme high temperature but differing fundamentally in their mechanisms—may reveal intriguing conclusions. Additionally, considering the changes in characteristics of these two types of CEs under different emission scenarios will aid in more accurately forecasting future climate risks.

When measuring CEs, it is crucial to select or construct relevant indicators. While duration and frequency of occurrence are commonly referenced, assessing the intensity of CEs is of greater importance. In previous studies, the severity assessment of CHTDE has predominantly relied on monthly-scale drought indices (e.g., Drought intensity index, Standardized precipitation evapotranspiration index) combined with monthly mean high-temperature metrics ([Wu and Jiang, 2022](#), [Sedlmeier et al., 2018](#)). However, this approach inherently limits the accurate characterization of their true severity. For instance, it fails to adequately capture the duration of extreme short-term (sub-monthly) CHTDE ([Yu and Zhai, 2020a](#), [Yu and Zhai, 2020b](#)). Additionally, classifying an entire month (or months) as a “drought month” may lead to either overestimation of event severity/duration or obscuration of short-lived, high-intensity extremes ([Li et al., 2023](#), [Jiang et al., 2025](#)). To overcome these limitations and more precisely quantify the intensity and destructive potential of CEs, this study introduces the concept of “magnitude index (MI)” This concept, based on daily-scale data, integrates the duration and intensity of CEs to provide a novel and refined severity classification method. It enables more effective intensity assessment, particularly excelling in capturing short-duration, high-intensity events and distinguishing the cumulative impacts of events with varying durations. This concept has not been widely applied in previous studies, representing an important supplement to CEs intensity assessment methodologies.

Given the complexity of CEs, accurately assessing their joint occurrence risk represents a critical challenge in current climate change research. In previous studies on evaluating the joint risk of CEs, copula functions have gradually been employed in the assessment of extreme climate events risks due to their unique dependence analysis characteristics. These applications include drought risk analysis ([AghaKouchak et al., 2014](#), [Varol et al.,](#)

2023), extreme heat and precipitation scenarios assessment (Tu et al., 2023, Liu et al., 2022), and standardized dry-heat compound indices construction (Li et al., 2021a, Li et al., 2021b, Hao et al., 2018a, Hao et al., 2018b, Wang et al., 2021a, Wang et al., 2021b, Wang et al., 2021c). However, these studies often exhibit a critical limitation: they typically employ either a single parameter or a univariate dependence structure to characterize the joint distribution of variables (e.g., using only one-dimensional or two-dimensional copula functions). This approach is insufficient for investigating more nuanced, multidimensional dependence variations and more complex tail dependence structures in CEs. Furthermore, the applications of this approach to CHTWE study remain relatively scarce. Investigating multidimensional models to assess the joint occurrence risks of such events is an urgent and necessary task in the field of compound climate events, particularly under future climate change conditions. The innovation and motivation of this study lie in not only applying the MI to the assessment of the intensity of CEs, but more importantly, we construct a three-dimensional copula model. This approach aims to overcome the dimensional and dependence structure limitations of traditional methods, thereby revealing more detailed and comprehensive characteristics of joint occurrence risks for both CHTDE and CHTWE across different dimensions, as well as their evolution under climate change.

Due to a warming rate that exceeds the global average (Hao, 2022), extreme rainfall events in China are also occurring at a rate nine times higher than the global average (Liu et al., 2024a, Liu et al., 2024b), China has become one of the regions most susceptible to frequent climate disasters worldwide. This study analyzed the total duration, maximum duration, and total frequency of occurrence of CHTDE/CHTWE in China under two emission scenarios (Shared Socioeconomic Pathways2-RCP4.5 and Shared Socioeconomic Pathways5-RCP8.5, SSP245 and SSP585) simulated by the Coupled Model Intercomparison Project Phase 6 (CMIP6), focusing on both historical periods (1982–2014) and the twenty-first century (2015–2100). Additionally, we introduced the concept of MI to assess the intensity of CEs. More importantly, we constructed one-dimensional, two-

dimensional, and three-dimensional copula models to evaluate the joint occurrence risks of CHTDE and CHTWE.

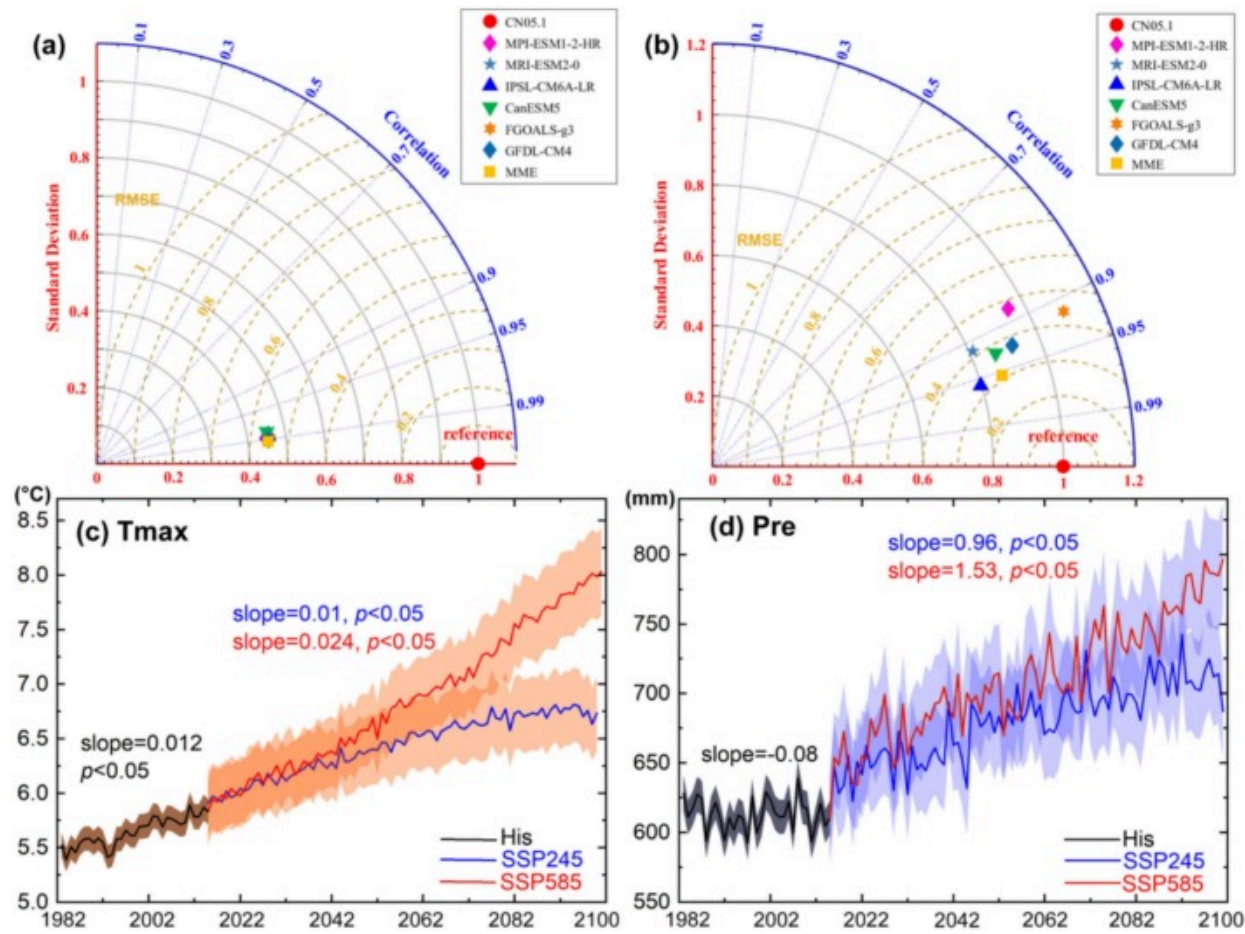
2. Materials and methods

2.1. Materials

This research evaluate CEs based on daily scale data, which avoids the rough identification caused by assessting CEs based on monthly scale data. The daily precipitation and daily maximum temperature data for China used in this study were derived from six models within the CMIP6 framework: CanESM5, FGOALS-g3, GFDL-CM4, IPSL-CM6A-LR, MPI-ESM1-2-HR, and MRI-ESM2-0 (Table S1). All data were generated by ensemble member r1i1p1f1, and the multi model ensemble (MME) dataset was produced based on their average. The observed reference for precipitation and temperature is provided by the China Meteorological Data Fusion (CMDF) gridded dataset. We applied the Equidistant Cumulative Distribution Function (ECDF) method to perform bias correction for the daily precipitation and maximum temperature outputs from the models (Jung, 2005). The period from 1982 to 2014 is considered as the historical baseline, while 2015 to 2100 is designated as the forecast period. We focused on shared socioeconomic pathways (SSP) characterized by high mitigation challenges and low adaptation challenges, specifically selected SSP245 (an upgrade of RCP4.5 scenario based on SSP2, which reaches a radiative forcing of 4.5W/m^2 by 2100) and SSP585 (an upgrade of RCP8.5 scenario based on SSP5, with SSP585 being the only SSP scenario capable of achieving a radiative forcing of 8.5W/m^2 by 2100). To assess the performance of the six CMIP6 models and the MME dataset, we utilized daily maximum temperature and precipitation data from CN05.1 (<https://ccrc.iap.ac.cn/resource> ↗, where “CN” represents the China domain and “.1” denotes an advanced version of CN05) for the period of 1982–2014 as validation for the CMIP6 models. This dataset was constructed from high-quality daily observational data on a $0.25^\circ \times 0.25^\circ$ grid based on observations from over 2,400 national meteorological stations across China and has been widely applied in regional

meteorological studies within China (Li et al., 2023, Qiao et al., 2024). All CMIP6 model data were bilinearly interpolated to match the resolution of the CN05.1 dataset.

The Taylor diagram (Fig. 1), based on historical daily temperature and precipitation data from CN05.1 for the period of 1982–2014, was used to evaluate the performance comparison between the six CMIP6 models and the MME against CN05.1 historical data. It provides three metrics to evaluate the performance, including the Centered Root Mean Square Error (RMSE), Standard Deviation (SD), and Correlation Coefficient (COR). The results indicate that the CMIP6-MME exhibits good performance in reproducing daily maximum temperature and precipitation (Fig. 1ab), previous studies have also confirmed the excellent simulation effect of MME (Zhou et al., 2025, Zhao et al., 2015). The performance of MME in simulating temperature is significantly higher than that of precipitation, which is manifested by having higher correlation and being closer to the center of RMSE. Based on the MME data, the increasing trend of temperature under the SSP585 scenario indicates that by the end of the 21st century, the overall daily maximum temperature in China will be approximately 2.4°C higher than that at the end of the 20th century (Fig. 1c). The slight decreasing trend in historical precipitation has reversed under the SSP scenarios, with a growth trend of 1.53mm/year in precipitation under the SSP585 scenario (Fig. 1d). Significant increases in both precipitation and temperature are more likely to contribute to the occurrence of CHTWE in the future.



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Fig. 1. Taylor diagram and changes in daily maximum temperature and precipitation for the CMIP6 performance evaluation. The evaluation of daily maximum temperature (a) and precipitation (b) is based on six models from the CMIP6 and the generated MME compared to the CN05.1 observed data. The changes in daily maximum temperature (c) and total precipitation (d) during the historical period (1982–2014) and two emission scenarios (SSP245 and SSP585) for the future period (2015–2100) are presented, historical are represented by a black line, while SSP245 and SSP585 are represented by blue and red

lines, respectively, the shaded area represents one standard deviation of the values. Better performance is indicated by an RMSE value close to 0, and COR and SD values close to 1. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

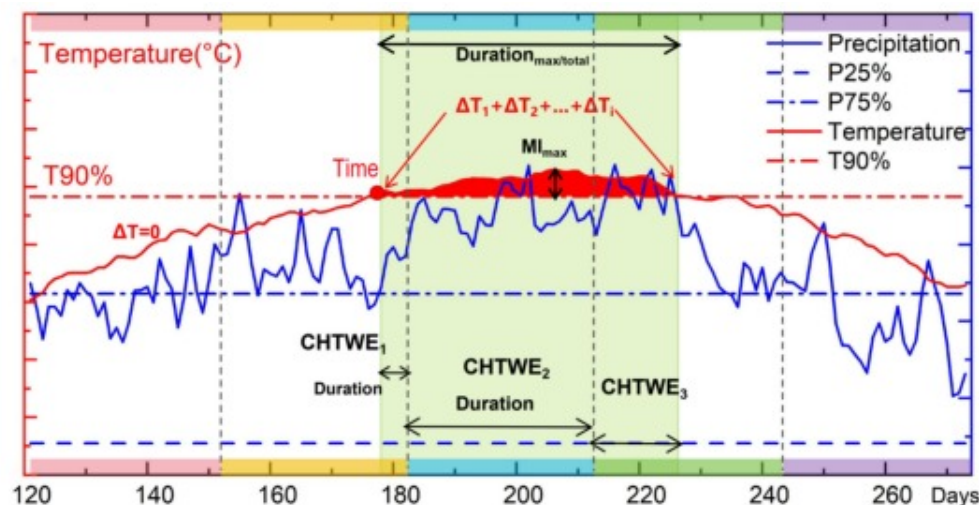
2.2. Methods

2.2.1. Definition of compound events

The prerequisite for defining an extreme event is to establish thresholds. Percentile thresholds are particularly suitable for large-scale climate assessments as they consider the actual meteorological conditions of different regions ([Robledo et al., 2020](#)).

Compound high temperature and dry or wet events are defined as instances where precipitation (P) falls below or exceeds a certain threshold, while the accompanying seasonal temperature (T) is above a specified threshold ([Wu and Jiang, 2022](#)). In this study, we define “high temperature” using the 90th percentile of daily maximum temperature (T90%), “dry” using the 25th percentile of daily precipitation (P25%), and “wet” using the 75th percentile of precipitation (P75%) ([Li et al., 2023](#), [Tencer et al., 2016](#), [Wu et al., 2019a](#)).

Initially, we iterated through daily maximum temperature and precipitation time series data from the historical period (1982–2014) to determine their percentile threshold layers. Considering spatial heterogeneity, we independently calculated the threshold for each $0.25^{\circ} \times 0.25^{\circ}$ grid cell, ensuring that the threshold is adapted to the local climatic background. Subsequently, we generated CEs for both historical and future phases by combining these threshold layers with actual data. We focused primarily on the duration, frequency, and magnitude index (monthly scale) of CEs, and further examined the total duration, maximum duration, total frequency, and annual mean magnitude of CEs at the annual scale. These are crucial attributes for understanding changes in extreme events. Their definitions are illustrated in [Fig. 2](#).



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Fig. 2. Schematic diagram of the definition method for compound events. The figure illustrates the precipitation and daily maximum temperature in China from May 1 to October 1, 2014, where the blue and red lines represent precipitation and daily maximum temperature, respectively. The blue and red straight lines indicate the 75th percentile threshold for precipitation and the 90th percentile threshold for temperature, while the blue dashed line represents the 25th percentile threshold for precipitation. The red filled area, which satisfies the conditions of both temperature and precipitation exceeding their respective thresholds, represents a CHTWE (across a period of three months). The difference between the top of the filled area and the threshold signifies the maximum MI, while the green shaded area indicates the longest duration of CHTWE observed during this period (If no other longer CHTWE occurred in that year, then this duration of occurrence would also be the total duration). The long bar shadows at the bottom represent different months. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

When calculating the MI, each event was normalized to ensure a uniform comparison, thereby placing its values within the range of [0, 1] (Russo et al., 2014). This normalization facilitates a comparison of temperature and precipitation conditions on the same scale. By multiplying these two normalized values, we obtained the MI of the CEs (Feng et al., 2020, Zhang et al., 2021b, Wang et al., 2021b, Zong et al., 2023). Compared to traditional methods for measuring the intensity of CEs, the MI calculation not only considers the intensity and duration of meteorological events but also quantifies and classifies the severity of the events. This provides a more detailed approach for cross-temporal and spatial comparisons, aiding in a more accurate assessment of the strength of CEs.

(1)

$$\text{CHTWE}_{\text{MI}} = P_{\text{MI}} \times T_{\text{MI}} = \left(R \sum_{i=1}^{D_i} \Delta T \right) \times \left(R \sum_{i=1}^{D_i} \Delta P \right), \quad \Delta P = P_i - P_{75\%} \quad P_i > P_{75\%} \quad \Delta T = T_i - T_{90\%} \quad T_i > T_{90\%}$$

(2)

$$\text{CHTDE}_{\text{MI}} = P_{\text{MI}} \times T_{\text{MI}} = \left(R \sum_{i=1}^{D_i} \Delta T \right) \times \left(R \sum_{i=1}^{D_i} \Delta P \right), \quad \Delta P = P_{25\%} - P_i \quad P_i < P_{25\%} \quad \Delta T = T_i - T_{90\%} \quad T_i > T_{90\%}$$

Let P_i be the normalized input data for the i -th range, and R be the normalized result. D_i represents the duration of extreme events, and we defined the differences between temperature and precipitation and their respective thresholds as ΔT and ΔP . Larger values indicate more severe conditions of high temperature and precipitation (Perkins-Kirkpatrick & Lewis, 2020). The variable ΔP can exist in two scenarios: above the 75% precipitation threshold and below the 25% precipitation threshold. The MI of the CEs was obtained by multiplying the MI of singular precipitation and temperature events. Furthermore, this study employed the least squares method to determine the linear trends of each indicator for both historical and future phases.

2.2.2. Copula return period calculation

We used two-dimensional copula models to assess the bivariate dependency between high temperature and precipitation. This approach can handle different types of

dependencies between variables and model them alongside their marginal distributions (Bevacqua et al., 2017, Li et al., 2015, Mazdiyasni et al., 2017). Additionally, we developed a three-dimensional copula based on three indicators of the two types of CEs to construct their return periods in both historical and future contexts. The return period refers to the average time interval expected for a certain extreme event to occur within a specified timeframe (Zscheischler and Seneviratne, 2017), which helps determine the rarity (or risk) of CEs (AghaKouchak et al., 2014, Zscheischler et al., 2020).

In this study, we employed the Generalized Extreme Value (GEV) distribution and Maximum Likelihood Estimation (MLE) methods to derive the return periods of extreme events from their statistical marginal distributions and the dependence structures between them. These two methods are considered beneficial for improving the accuracy of estimating the probability of CEs (Ridder et al., 2020, Zscheischler and Seneviratne, 2017).

2.2.2.1. Generalized extreme value distribution

Before evaluating the two-dimensional copula model, we extracted daily maximum temperature and precipitation data from historical records and two future simulation scenarios. The GEV was employed to fit the marginal distributions of these indicators, aiming to more accurately describe their extreme behavior (Coles et al., 2001), and subsequently calculate the return periods for temperature and precipitation (Zhang et al., 2022a, Das and Allen, 2024). The general form is:

$$F(x; \mu, \sigma, \xi) = \begin{cases} \exp\left[-\left(1 + \xi \frac{x-\mu}{\sigma}\right)^{-1/\xi}\right], & \text{if } \xi \neq 0 \\ \exp\left[-\exp\left(-\frac{x-\mu}{\sigma}\right)\right], & \text{if } \xi = 0 \end{cases} \quad (3)$$

where $F(x; \mu, \sigma, \xi)$ is the cumulative distribution function of the GEV distribution; μ is the location parameter, which represents the center of the distribution; σ is the scale parameter, which controls the dispersion of the distribution; ξ is the shape parameter, which determines the type of distribution (Bali, 2003). For the GEV distribution, the

extreme value x_T corresponding to a specific return level T can be calculated using the inverse cumulative distribution function (Inverse CDF) (Huang et al., 2016):

$$x_T = \begin{cases} \mu + \frac{\sigma}{\xi} \left[\left(-\ln \left(1 - \frac{1}{T} \right) \right)^{-\xi} - 1 \right], \xi \neq 0 \\ \mu + \sigma \ln \left(-\ln \left(1 - \frac{1}{T} \right) \right), \xi = 0 \end{cases} \quad (4)$$

2.2.2.2. Two-dimensional copula

The analysis of the two-dimensional copula requires the extraction of daily maximum temperature and precipitation data under both historical and future scenarios, as well as the estimation of their marginal distributions. We employed the normal distribution, GEV distribution, gamma distribution, and log-normal distribution for fitting to determine the optimal marginal distribution.

As mentioned in Section 2.2.1, CEs are defined as instances where precipitation and maximum temperature are above or below a certain threshold (x, y) . Therefore, the general forms of the marginal distribution functions $F_X(x)$ and $F_Y(y)$ can be expressed as $P(X \leq x)$ and $P(Y \geq y)$, respectively. The joint distribution function is given by:

$$P(X \leq x, Y \geq y) = C(F_X(x), F_Y(y)) = C(u, v) \quad (5)$$

here, $F_X(x)$ and $F_Y(y)$ denote the cumulative distribution functions for variables X and Y (Ayantobo et al., 2018), which in this study correspond to daily maximum temperature and daily precipitation, respectively; P represents the probability, and C denotes the joint distribution function of the two variables, that is, the copula model. The return period T can be represented by:

$$T = \frac{1}{P(X \leq x, Y \geq y)} \quad (6)$$

$$T = P(X \geq x, Y \leq y) = 1 - C(F_X(x), F_Y(y)) \quad (7)$$

We utilized different types of copulas, including Gaussian copula, t copula, Frank copula, Gumbel copula, and Clayton copula (Nelsen, 2006, Bevacqua et al., 2019).

Gaussian Copula:

$$C(u, \nu) = \Phi_2(\Phi^{-1}(u), \Phi^{-1}(\nu); \rho) \quad (8)$$

t Copula:

$$C(u, \nu) = t_n(t^{-1}(u), t^{-1}(\nu); \rho, \nu) \quad (9)$$

Frank Copula:

$$C(u, \nu) = -\frac{1}{\theta} \ln(1 + (e^{-\theta u} - 1)(e^{-\theta \nu} - 1) / (e^{-\theta} - 1)) \quad (10)$$

Gumbel Copula:

$$C(u, \nu) = \exp(-[(-\ln u)^\theta + (-\ln \nu)^\theta]^{1/\theta}) \quad (11)$$

Clayton Copula:

$$C(u, \nu) = \max((u^{-\theta} + \nu^{-\theta} - 1)^{-1/\theta}, 0) \quad (12)$$

where Φ represents the CDF of the standard normal distribution, while Φ^{-1} denotes the inverse CDF of the standard normal distribution. Φ_2 represents the correlation coefficient matrix (covariance matrix) of the two vectors u and ν . T_n is the CDF of the t-distribution with n degrees of freedom, and T^{-1} signifies the inverse CDF of the n -distribution. In the Frank copula, Gumbel copula, and Clayton copula, θ is the shape parameter that controls the strength of dependence. When $\theta > 0$, it indicates positive dependence, while $\theta < 0$ signifies negative dependence.

The optimal copula is selected by comparing the AIC and BIC values, as well as the squared Euclidean distance, of different copulas. Additionally, the parameters of these bivariate copulas were estimated using the maximum likelihood estimation method (Akaike, 2011).

$$\text{AIC} = -2\log(L) + 2k \quad (13)$$

$$\text{BIC} = -2\log(L) + k\log(n) \quad (14)$$

where L is the likelihood function, k is the number of parameters, and n is the sample size. After analysis, the Gaussian copula was selected as the optimal copula ([Table S2](#)).

2.2.2.3. Three-dimensional copula

Consistent with the two-dimensional copula function, it is first necessary to compute the cumulative distribution function using the selected marginal distributions. In the three-dimensional copula, the variables represent the duration, frequency, and magnitude of composite events.

$$\text{EP}_i = \text{FX}_i(\mathbf{x}) \quad (i \in \{1, 2, 3\}) \quad (15)$$

where $i=1,2,3$ corresponds to the CDFs of the three indicators.

Unlike the two-dimensional copula, for three-dimensional data, we employed the Vine copulas to determine the optimal structure and parameters. Compared to the traditional copula functions, the Vine copulas can decompose multidimensional dependence structures into a series of bivariate copula functions. This hierarchical structure allows for multivariate modeling, capturing more complex nonlinear and asymmetric dependencies among high-dimensional variables while significantly reducing the complexity of high-dimensional parameter estimation ([Bedford and Cooke, 2002](#), [Zeng et al., 2022](#)). The Vine copulas serve as an effective tool for integrating different bivariate distributions and computing multivariate conditional probabilities ([Ni et al., 2020](#)). The joint probability distribution of the three bivariate copulas and the Vine copula formula are as follows:

$$\text{EP}_{12} = C(U_1, U_2 | \text{Family1}, \theta_1) \quad (16)$$

$$\text{EP}_{13} = C(U_1, U_3 | \text{Family2}, \theta_2) \quad (17)$$

$$EP_{23} = C(U_2, U_3 | \text{Family3}, \theta_3) \quad (18)$$

$$C(U_1, U_2, U_3) = \text{vinecdf}(U_1, U_2, U_3 | A, \text{fam}, \theta) \quad (19)$$

where A represents the structure of the Vine copula, fam denotes the selected optimal copula family, which is typically determined using statistical model selection criteria such as AIC or BIC, and θ refers to the parameters associated with each copula, which adjust the degree of dependence within the copula.

We calculated the joint return periods T_{joint} /probabilities P_{joint} , co-occurrence return periods T_{CO} /probabilities P_{CO} , and conditional return periods $T_{\text{condition}}$ /probabilities $P_{\text{condition}}$ for the three indicators of CHTDE and CHTWE at different stages. The results for the co-occurrence return periods are presented in the main text. The joint return period among the three indicators was obtained through the joint CDF, which refers to the average time interval during which the random variables are observed to exceed a certain threshold simultaneously. The calculation formulas are as follows (Liu et al., 2015):

$$P_{\text{joint}} = 1 - C(U_1, U_2, U_3) \quad (20)$$

$$T_{\text{joint}} = \frac{1}{1 - C(U_1, U_2, U_3)} \quad (21)$$

The co-occurrence probability and co-occurrence return period (the average time interval in which the variables are observed to exceed their respective thresholds at least once within a specified time) are derived from the following formulas (Tao et al., 2021):

$$P_{\text{co}} = 1 - U_1 - U_2 - U_3 + P_{\text{dp}} + P_{\text{dv}} + P_{\text{pv}} - C(U_1, U_2, U_3) \quad (22)$$

$$T_{\text{co}} = \frac{1}{1 - U_1 - U_2 - U_3 + P_{\text{dp}} + P_{\text{dv}} + P_{\text{pv}} - C(U_1, U_2, U_3)} \quad (23)$$

The calculation methods for conditional probability and conditional return period (the average time interval in which additional variables exceed their thresholds given that one

variable exceeds its threshold) are as follows (He et al., 2024):

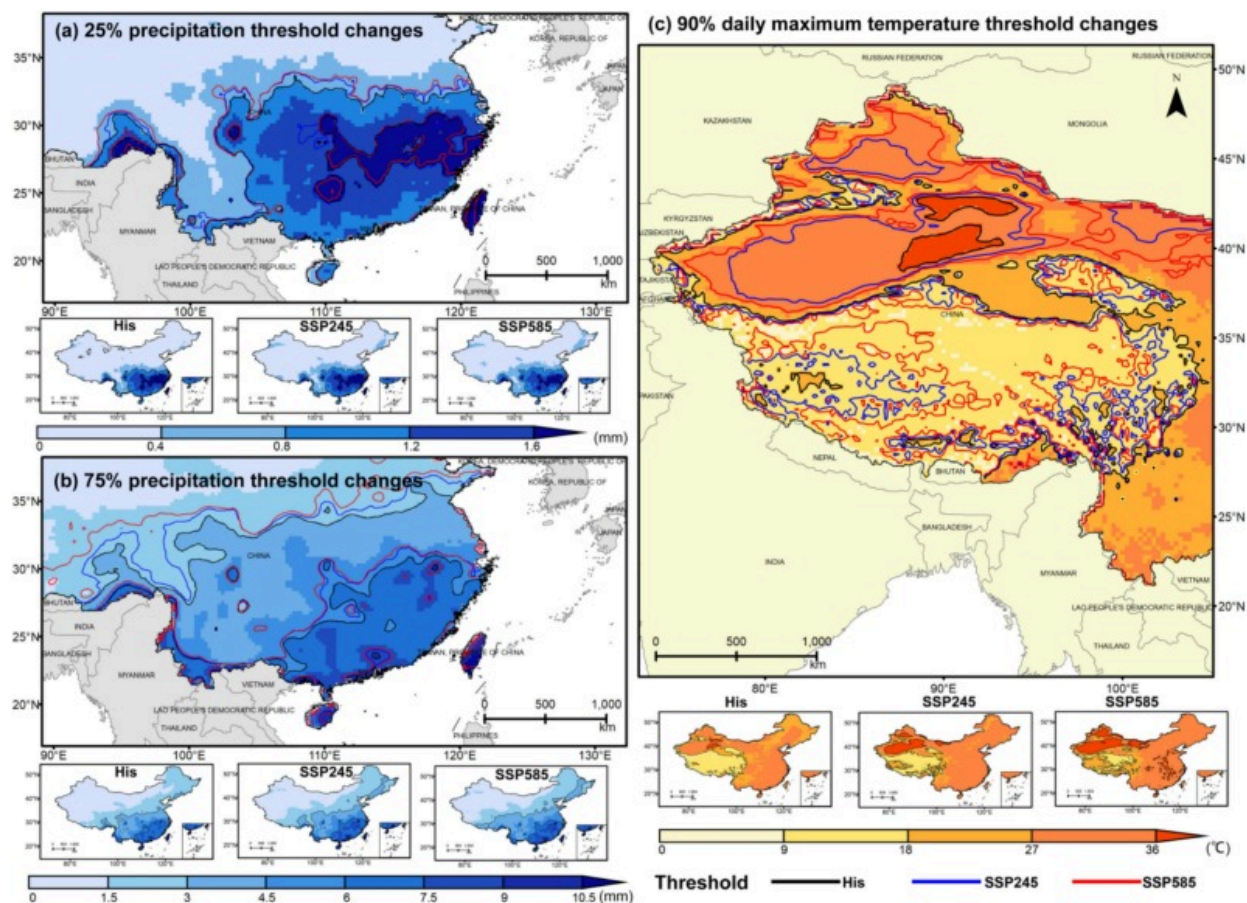
$$P_{\text{condition}} = P(U_1, U_2 | U_3) = \frac{C(U_1, U_2, U_3)}{P(U_3)} \quad (24)$$

$$T_{\text{condition}} = \frac{1}{1 - \frac{C(U_1, U_2, U_3)}{P(U_3)}} \quad (25)$$

3. Results

3.1. Expansion of historical and future threshold spatial ranges

Given that the definition of CEs is inherently dependent on thresholds, and to facilitate a comprehensive understanding of how these events evolve under climate change, we compared the temperature and precipitation thresholds between the historical period (1982–2014) and the last 33 years of the 21st century (2068–2100) (Fig. 3). The results indicated that the precipitation threshold lines shifted northward overall from the southern regions. A 10% reduction in the spatial extent of the 25% and 75% precipitation threshold isolines signified a decrease in low precipitation areas (with $P_{25\%} < 0.4\text{mm/day}$ and $P_{75\%} < 1.5\text{mm/day}$; Fig. 3ab).



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Fig. 3. Spatial distribution of threshold changes. Against the backdrop of the threshold layers from the historical period (1982–2014), the spatial distribution of the 25% precipitation threshold (a), the 75% precipitation threshold (b), and the 90% daily maximum temperature threshold (c) is shown in the enlarged sections. Black represents the historical period, while blue and red represent the distribution of threshold contour lines for SSP245 and SSP585 scenarios, respectively. The three small maps below each large map also display the spatial distribution of thresholds for the historical period,

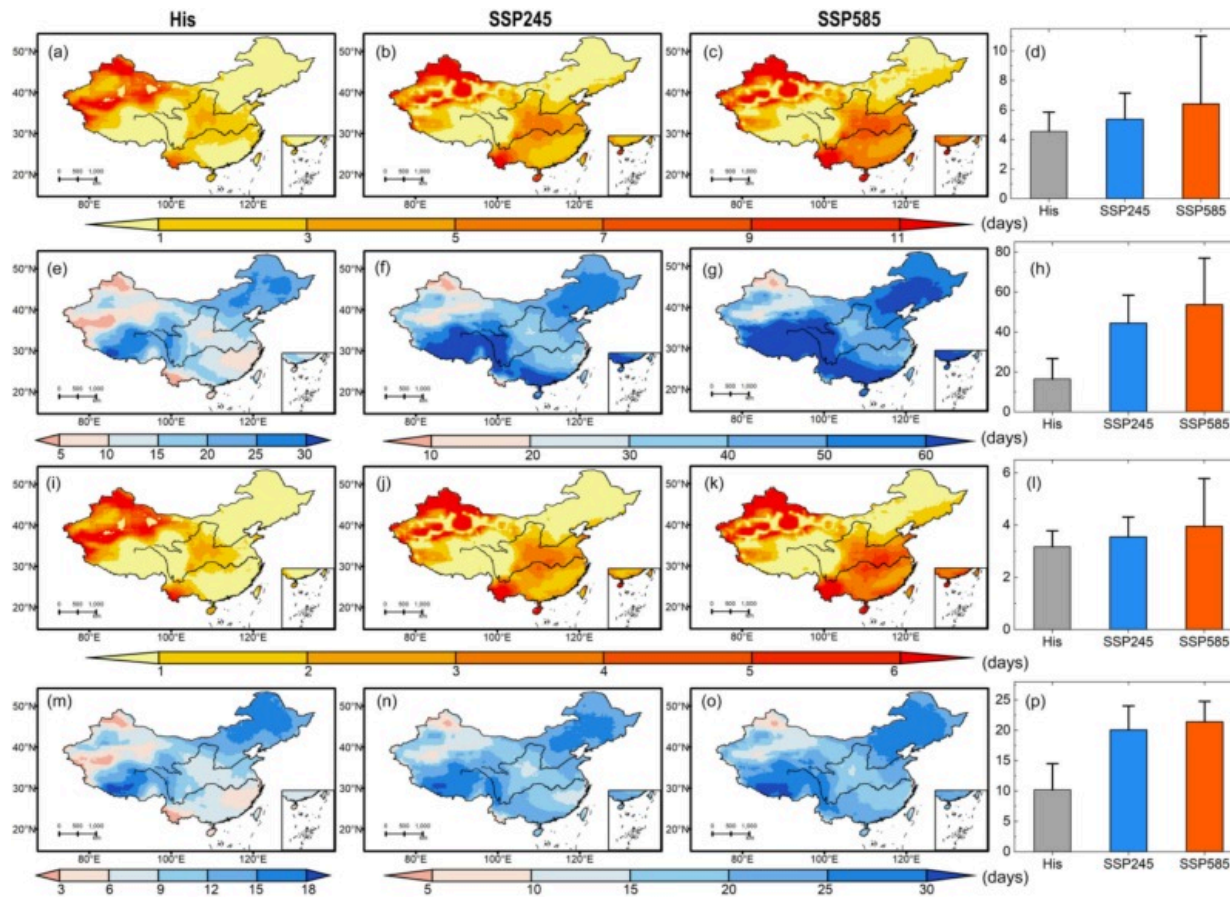
SSP245, and SSP585 scenarios. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

In contrast to the northward shift of the precipitation isolines, the changes in temperature thresholds are featured as an expansion into surrounding areas, with the spatial extent of temperatures exceeding 36°C increased by 15%, particularly noticeable in the Xinjiang region. Compared to the observed period, the 90% temperature threshold in the last 33 years under the SSP585 scenario increased by 5.2°C. The changes in precipitation thresholds were not particularly pronounced (with an increase rate of 0.06mm/day and 0.5mm/day for the 25% and 75% precipitation thresholds, respectively), linking this to the notable growth in total precipitation presented in [Fig. 1d](#), this discrepancy might indicate a signal of prolonged individual rainfall events. Furthermore, these results suggest that percentile thresholds defined for the historical period could become quite common in the future, thereby validating the general intensification of extreme values due to global warming.

3.2. Characteristics of compound events in China: Historical and future

3.2.1. Spatial characteristics

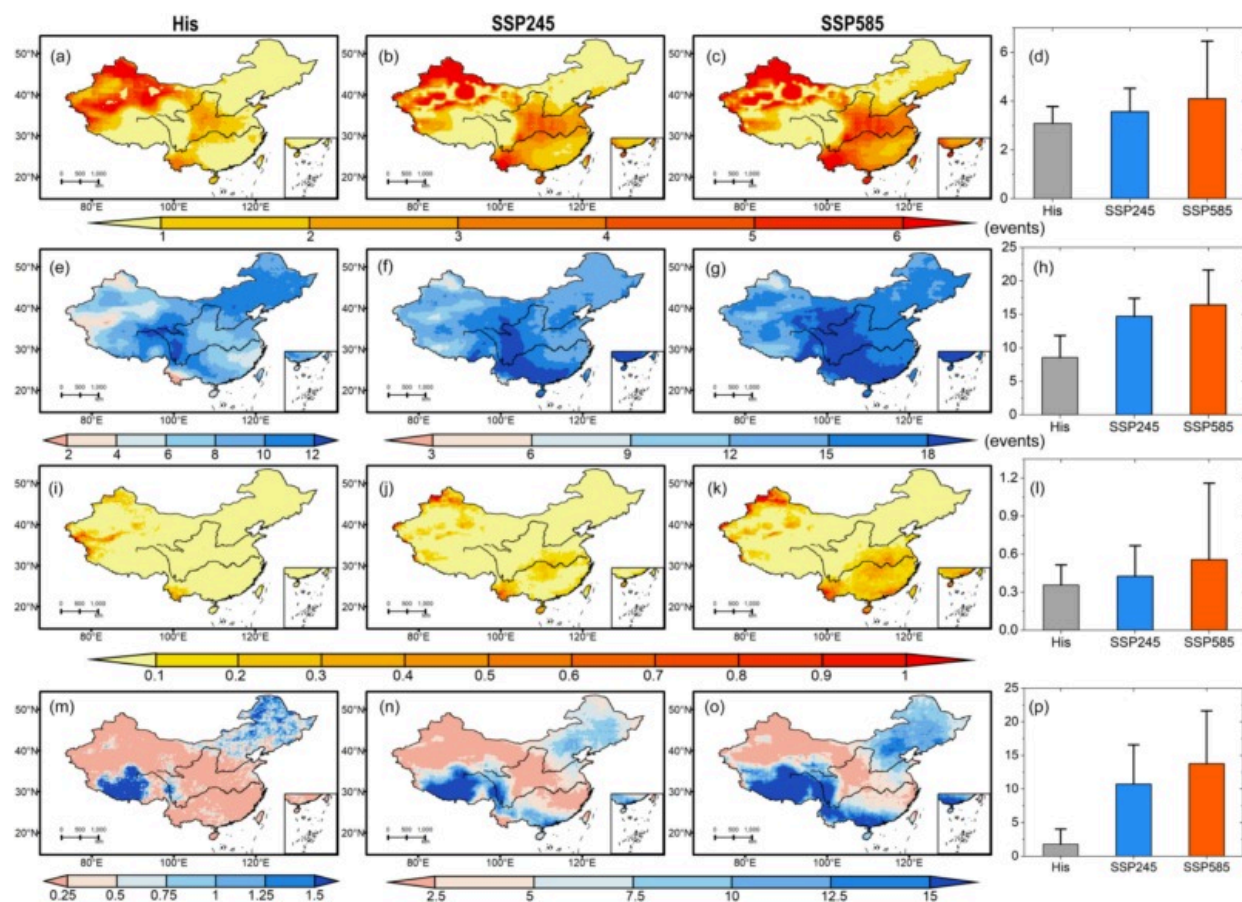
We compared the characteristics of CHTDE and CHTWE under the historical (His) and two future scenarios (SSP245 and SSP585) ([Fig. 4](#), [Fig. 5](#)); the overall mean and increases in CHTDE under the two emission scenarios were significantly lower than those of CHTWE. The total duration of CHTDE and CHTWE ([Fig. 4a-c](#), [e-g](#)) ranged from 2.47 to 5.85days/year and from 6.77 to 26.64days/year, respectively. The maximum duration fluctuations ([Fig. 4i-k](#), [m-o](#)) ranged from 2.01 to 3.79days/year for CHTDE and from 5.33 to 14.49days/year for CHTWE. The frequency ([Fig. 5a-c](#), [e-g](#)) varied from 1.89 to 3.78 times/year for CHTDE and from 4.55 to 11.84 times/year for CHTWE, while the MI ([Fig. 4i-k](#), [m-o](#)) ranged from 0.17 to 0.57 and from 0.47 to 3.8, respectively.



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Fig. 4. Spatial distribution of total/maximum duration of CHTDE and CHTWE. The spatial distribution and multi-year averages of the total duration of CHTDE during the historical period (a), SSP245 (b), and SSP585 (c) are presented in the first row. Correspondingly, the second row (e-h) displays the spatial distribution and averages of the total duration of CHTWE. The third and fourth rows show the spatial distribution and average magnitude of CHTDE (i-l) and CHTWE (m-p) under different scenarios, respectively.



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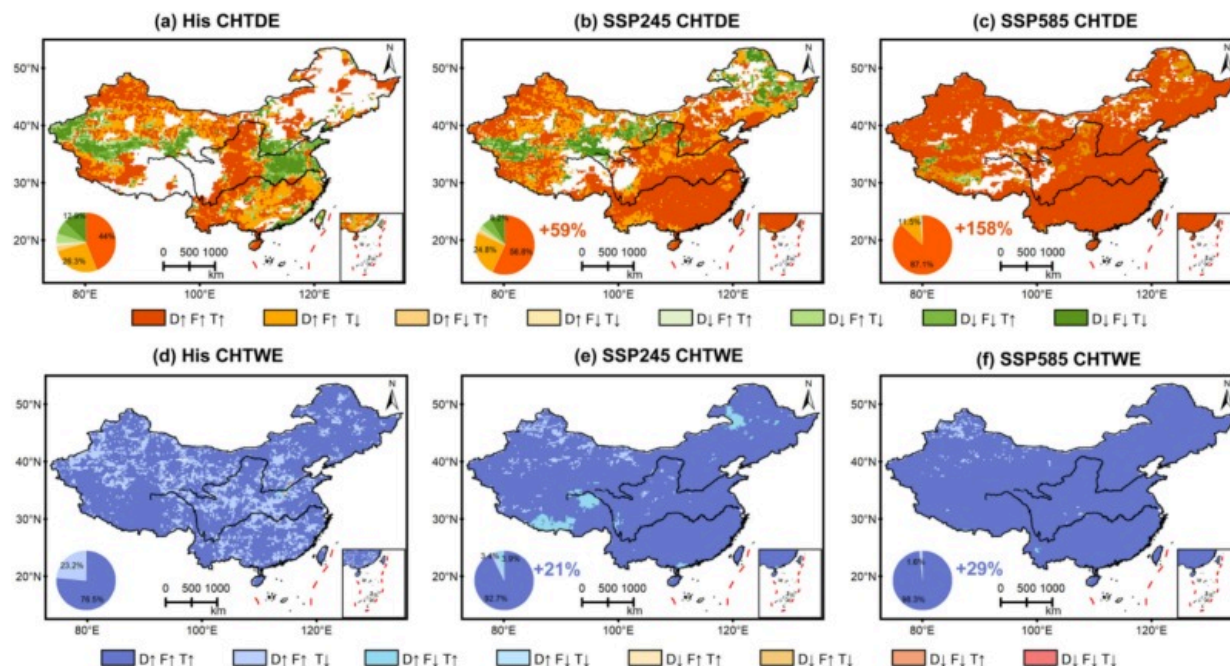
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Fig. 5. Spatial distribution of total frequency and annual mean MI of CHTDE and CHTWE. The occurrence frequency of CHTDE during the historical period (a), SSP245 (b), and SSP585 (c) is presented in the first row, along with the multi-year averages (d). Correspondingly, second row (e-h) shows the spatial distribution and averages of the frequency of CHTWE. The third and fourth rows illustrate the spatial distribution and average magnitude of CHTDE (i-l) and CHTWE (m-p) under different scenarios, respectively.

Compared to the historical conditions, under the higher emission scenario (SSP585), the increase in the four indicators of CHTDE ranged from 25% to 55%, while the increase for CHTWE ranged from 92% to 677%. In future, if emissions were controlled to a moderate scenario (SSP245), the occurrence of CHTDE is expected to decrease by 11.5% to 30%, and for CHTWE, it would decrease by 7% to 28% (compared to SSP585).

In terms of spatial distribution, CHTDE is more concentrated in Xinjiang and Yunnan provinces of China. With increasing emissions, the rise of such events is particularly pronounced in central China. The Tibetan Plateau and Northeast China were the primary regions for the distribution and increase of CHTWE, which will further expand to the southern coastal areas. This region is expected to experience longer durations, more frequent occurrences, and greater intensities of CHTWE in the future.

Multiyear trend analysis indicated that the total duration of both types of CEs increased rapidly. If emissions were controlled under the SSP245 scenario, the growth trend of total duration for CHTDE is expected to be reduced by a factor of three ([Fig. S1](#)); we assessed the overall changes by overlaying and combining the trends of various indicators ([Fig. 6](#)). Results from the historical period of 1982–2014 showed that 44% of China's land surface experienced more, larger, and intenser CHTDE. Following this, there were more persistent but less intense CEs, with regions experiencing decreases in frequency, duration, and MI accounting for 12% ([Fig. 6a](#)). Projections for SSP245 and SSP585 suggest that this observed trend will continue in the future, 54% and 87% respectively of areas are expected to experience more, larger, and intenser CHTDE between 2015 and 2100 ([Fig. 6b,c](#)). In the historical period, the multi-index collaborative growth regions of CHTWE occupied 76.2% of the spatial extent ([Fig. 6d](#)), and this is projected to further increase by 21% and 29% under the SSP245 and SSP585 scenarios, respectively ([Fig. 6e,f](#)).



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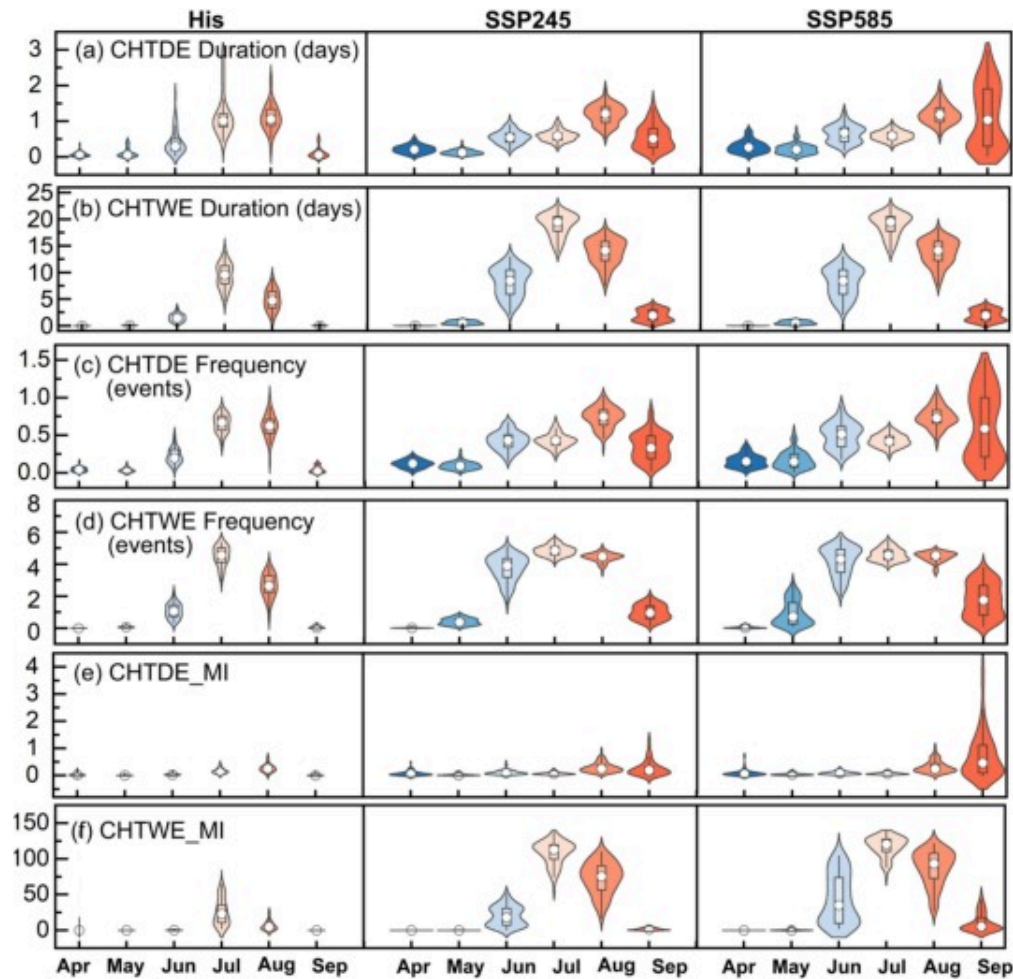
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Fig. 6. Overlay analysis of compound event trends. Based on the multi-year trends of CEs' duration, frequency, and MI, an overlay analysis was conducted to generate eight scenarios (see legend). The overlay results for compound hot dry events under historical (a), SSP245 (b), and SSP585 (c) scenarios are presented in the first row, while the second row (d-f) displays the spatial overlay distribution of CHTWE under different scenarios. In the legend, D represents duration, F represents frequency, and T represents intensity, with upward and downward arrows indicating increasing and decreasing trends, respectively, while the pie charts represent the proportions of the eight scenarios.

3.2.2. Monthly characteristics

In addition to variations on annual and spatial scales, we also examined the occurrence of CEs on monthly scale (Fig. 7). Both types of events are concentrated in July and August;

however, with future climate change, the peak of CHTDE shifted to August/September under the SSP585 scenario, showing a trend of spreading into the spring months (April/May). Compared to the SSP585 scenario, if emissions were controlled under the SSP245 scenario, the duration, frequency, and MI of CHTDE in spring (March/April/May) could be reduced by 62%, 60%, and 54%, respectively.



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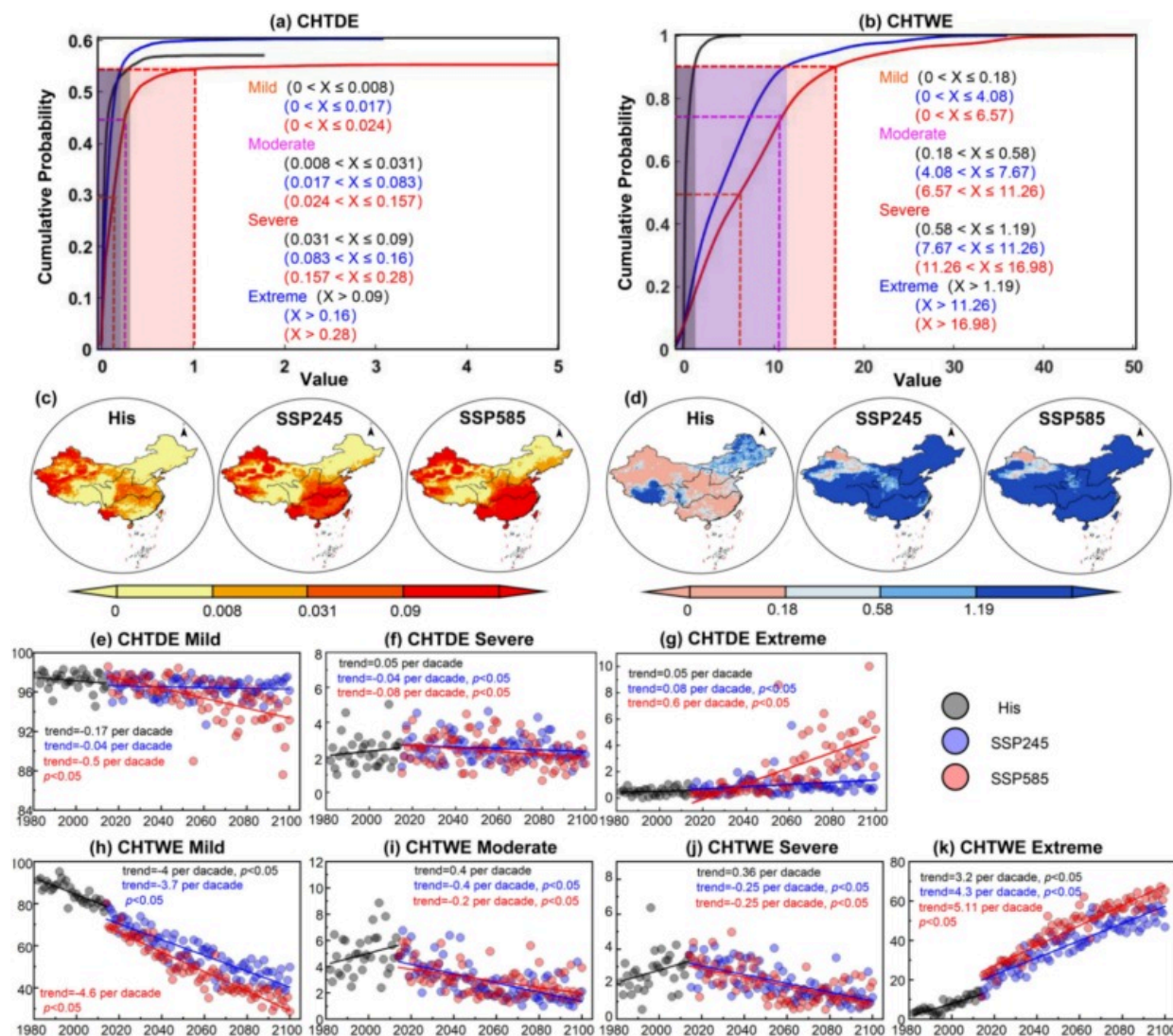
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Fig. 7. Monthly distribution of CEs. Each small box displays the distribution of CHTDE' duration (a), frequency (c), and MI (e) from April to September. Similarly, the monthly distribution of CHTWE can be seen in panels b, d & f. We also differentiated both types of events across the three periods (His, SSP245, and SSP585).

CHTDE occurred over a broader range of months than CHTWE, but their occurrence duration, frequency, and MI were all shorter than those of CHTWE. In contrast, CHTWE remains concentrated in July and August, with a more “uniform” distribution across these months. Similar to multi-year trends (Fig. 4, Fig. 5), the increases in CHTWE under both future scenarios were not substantial; however, compared to historical periods, CHTWE showed significant increases in June, July, August, and September.

3.3. Changes in the magnitude index of compound events

By extracting the MI of CHTDE and CHTWE from 1982 to 2014, we plotted cumulative probability density function curves (Figs. 8a, b). Three thresholds (set at cumulative probabilities of 0.5, 0.75, and 0.9) were established to classify events into four levels: mild, moderate, severe, and extreme (Martínez-Lozano et al., 2002, Russo et al., 2014). As emissions increase, CEs become more severe and persistent (Figs. 8a, b).



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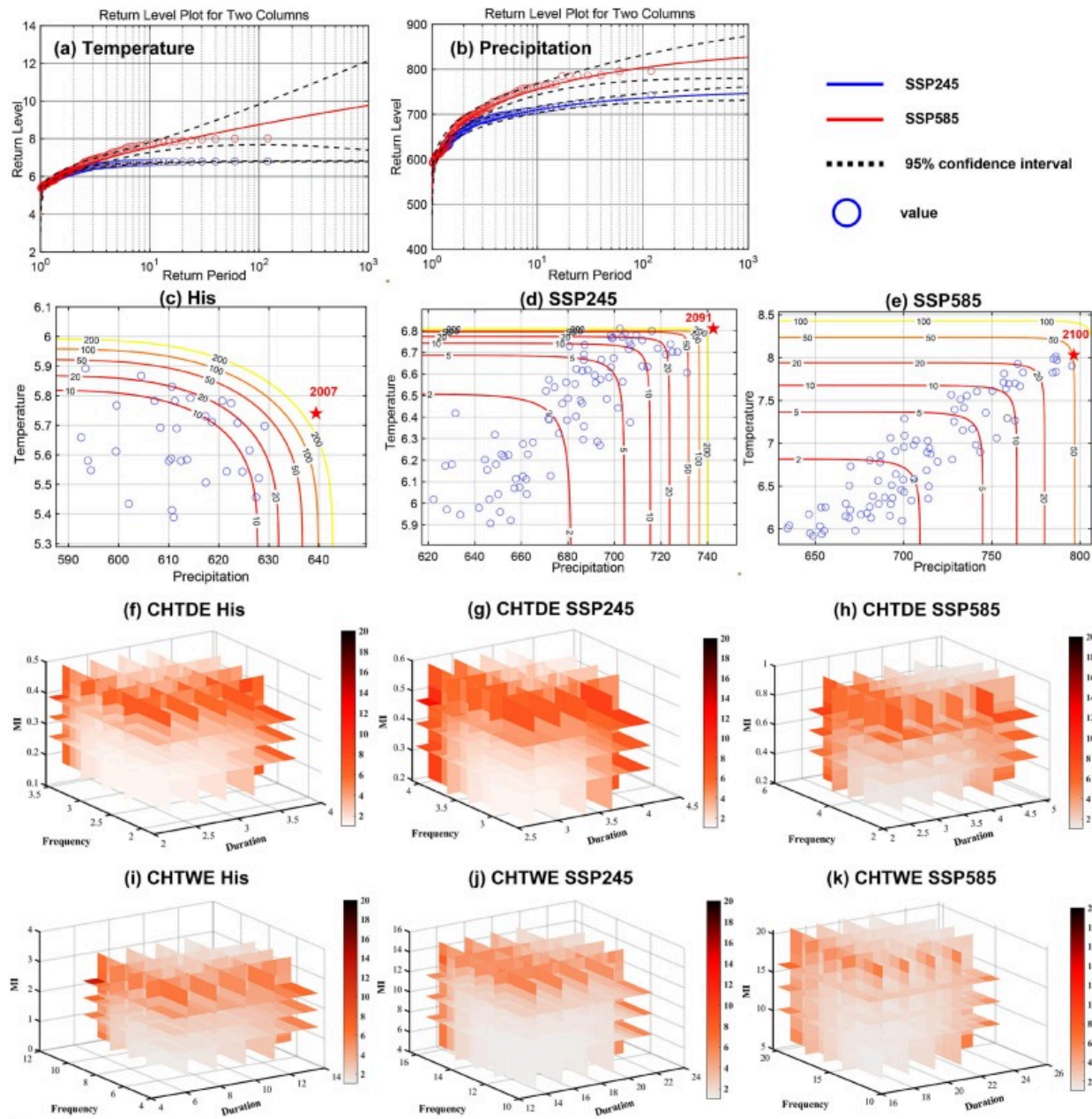
Fig. 8. MI characteristics of compound events. The cumulative probability distributions of MI for CHTDE and CHTWE are shown in panels a and b, the black, blue, and red lines represent the past, SSP245, and SSP585, respectively. The spatial distribution of CHTDE and CHTWE based on the cumulative probability density function defined for four

magnitude levels during the three periods (historical, SSP245, and SSP585) are shown in panels c and d. The multi-year scatter distributions of mild, severe, and extreme of CHTDE are display in panels e-g, while panels h-k represent the MI of mild, moderate, severe, and extreme of CHTWE. The horizontal axis represents the years, while the vertical axis shows the proportion of earthquakes of a specific magnitude in that year. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Based on the spatial representation of multi-year MI of CHTDE and CHTWE (Figs. 8c, d), it can be observed that extreme dry conditions nearly cover the entire southern region of China, while CHTWE has spread to all areas beyond Xinjiang. In addition to overall changes and spatial distributions, we separately displayed the MI characteristics for the four levels (Fig. 8e-k). For both CHTDE and CHTWE, the growth of extreme events was particularly significant (Figs. 8g, k, all passing the $p < 0.05$ test), primarily driven by transitions from mild events (Figs. 8e, h).

3.4. Joint occurrence risks of compound events

We investigated the return periods of temperature and precipitation based on GEV and implemented bivariate and trivariate copula models to jointly reproduce the indicators of the combination of temperature and precipitation, as well as CEs in both historical and future scenarios. The return periods for temperature and precipitation revealed a significant upward trend in both scenarios (Figs. 9d, e), indicating more frequent of CHTWE.



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Fig. 9. Results of the copula model. The return periods for temperature and precipitation are displayed in panels a-b. The two dotted lines above and below the solid line represent the 95% upper and lower confidence intervals. The joint distribution and return periods of temperature and precipitation under the historical baseline, SSP245, and SSP585 scenarios are presented in panels c-e. The three-dimensional histograms illustrate the co-occurrence return periods between the duration, frequency, and magnitude of CHTDE (f-h) and CHTWE (i-k), the numbers in the legend represent the return periods. The joint return period is a key indicator for measuring the severity of CEs. A larger value indicates a lower probability of occurrence for the CEs, and thus higher extremity (severity).

Following a two-dimensional Copula joint return period analysis of historical simulated data from 1982 to 2014, we identified the CEs with the longest joint return period during this period to have occurred in 2007 (Fig. 9a). During that year, China experienced its worst drought in a decade, while significant heavy rainfall events were also observed in both southern and northern regions (Liu et al., 2024a, Liu et al., 2024b;). However, under the SSP245 and SSP585 scenarios, the likelihood of annual precipitation and maximum temperature exceeding the intensity of 2007 was estimated at 93% and 97%, respectively (Figs. 9b, c). As emissions increase, the combination of precipitation and temperature trended toward extreme conditions, with the return periods of CEs rapidly shortened. Under the SSP585 scenario, between 2015 and 2100, there will be no more than a 50-year return period for CEs.

We constructed the return periods for the duration, frequency, and MI of CHTDE and CHTWE using a three-dimensional copula model. Under the historical conditions, the frequency and duration of CEs exhibited lower interaction intensity (Figs. 9f, i); however, with the increase of emissions (Fig. 9g-h, j-k), under the same return period, the values of all three indicators enhanced continuously, this implies that CEs will become more frequent and severe, especially CHTWE. Furthermore, for the same year, the three-dimensional joint return period considering the duration, frequency, and MI of CEs was consistently shorter than the two-dimensional joint return period for temperature and precipitation, and it was also shorter than the univariate return periods. This implies that

incorporating the variables of CEs provides greater practical significance for assessing the risks associated with historical and future CEs. Overall, for both types of CEs, the results from all dimensions captured the increase in event frequency and the shortening of return periods under future emission scenarios.

4. Discussion

4.1. More frequent and serious compound events

In recent years, discussions on compound events have become increasingly widespread. The primary focus of this study's results is: the spatial distribution and growth trends of CHTDE and CHTRE. The high incidence of CHTDE in Xinjiang, Yunnan, and the middle and lower reaches of the Yangtze River are inline with the results of previous studies ([Zhou et al., 2025](#), [Wu et al., 2021d](#), [Zou and Song, 2024](#), [Zhang et al., 2022b](#)). The growth of CHTWE is more intense than imagined, concentrated in the Tibetan Plateau ([Ning et al., 2022](#), [Li et al., 2023](#)), which may be related to the amplification effects of the Tibetan Plateau's elevation feedback and snow-albedo feedback, glacier and permafrost melt, thermal effects, and convection ([Wu et al., 2015](#), [Yao et al., 2015](#), [Wu et al., 2019a](#)). Compared to historical periods, the growth rates of all four indicators of CHTWE under the SSP585 scenario far exceed those of CHTDE ([Fig. 4](#), [Fig. 5](#)). Unlike the expansion of CHTDE into the spring season ([Fig. 7](#), [Yuan et al., 2024](#)), CHTWE were trending toward an overall enhancement in event intensity, which was another noteworthy phenomenon—the increase in MI ([Fig. 8](#)). In future scenarios, CHTDE are projected to increase significantly in southern China ([Fig. 8](#)), primarily driven by rising temperatures ([Fig. 3](#)). The growth rate for CHTWE is observed to be 5.11 per decade, indicating that by the end of the 21st century, CHTWE may become the predominant CEs in most regions of China, except for Xinjiang. Similar phenomena were noted in previous studies, which found that under global warming, the intensity and frequency of future wet-heat waves will further increase ([Speizer et al., 2022](#); [Coffel et al., 2017](#); [Wang et al., 2021a](#), [Wang et al., 2021b](#), [Wang et al., 2021c](#)). Compared to CHTDE, CHTWE will be more frequent, stronger, and last

longer in future ([Dunne et al., 2013](#), [Li et al., 2020a](#), [Li et al., 2020b](#), [Parsons et al., 2022](#), [Wang et al., 2021a](#), [Wang et al., 2021b](#), [Wang et al., 2021c](#), [Tang et al., 2025](#)), emphasizing the necessity of focusing on CHTWE.

In comparison to historical studies, our study employs Vine Copula functions to construct the dependency structure among the indicators of CEs (duration-frequency-MI), thereby enabling the analysis of their return periods building upon the assessment of CEs occurrence characteristics. The results showed that under high emission scenarios, as the frequency, duration, and MI of both types of CEs increased, the return periods shortened. This implies that high-frequency, persistent, and intense CEs will become more common in the future. While previous studies have attempted to analyze the joint distribution of multiple disaster-inducing factors using high-dimensional symmetric copula functions ([Xu et al., 2018](#), [He et al., 2021](#)), this approach has not yet been applied in the field of compound extreme climate events, thereby the present study deepened the understanding of the inherent interactions within these CEs.

4.2. The occurrence mechanism of CHTDE and CHTWE

Although CHTDE and CHTWE are both associated with high temperatures, their mechanisms of formation differ significantly. Land-atmosphere coupling is a key driving factor behind the increased severity and frequency of these CEs ([Shi et al., 2021](#)). CHTDE are triggered by similar atmospheric conditions, such as high-pressure systems, where the atmospheric conditions can mutually reinforce each other ([Seneviratne et al., 2010](#)). First, high-temperature phenomena exacerbate drought primarily through sensible and latent heat processes: In terms of sensible heat, greater incoming radiation can lead to the accumulation of sensible heat in the atmosphere ([Miralles et al., 2019](#)), contributing to heatwaves and potentially resulting in higher soil temperatures. Regarding latent heat, elevated temperatures can cause increased evaporation rates, which may create drier conditions ([Seneviratne et al., 2010](#)). At the same time, dry conditions are typically accompanied by reduced evaporation ([Trenberth et al., 2013](#), [Savenije, 2004](#)), decreasing the likelihood of precipitation. Finally, the potential positive feedback between drought

and heatwaves helps retain warm air in the atmospheric boundary layer, leading to more intense and prolonged heatwaves ([Miralles et al., 2014](#)). Heatwaves can act both as a driver of drought occurrence and as a consequence of drought conditions.

Moreover, elevated surface temperatures and sensible heat flux associated with heatwaves can enhance convective available potential energy in the atmosphere (unstable energy), indicating a greater potential for instability and stronger convective activity, such as severe precipitation events ([Zhang and Villarini, 2020](#), [Wu et al., 2021b](#), [Yin and Slater, 2023](#), [Ning et al., 2022](#), [Hao, 2022](#)). The alternating presence of anomalous anticyclones and cyclones can produce sustained high temperatures and extreme precipitation ([Chen and Zhai, 2017](#), [Hsu et al., 2016](#), [Liao et al., 2021](#), [Ning et al., 2022](#), [You and Wang, 2021](#)). Additionally, the atmospheric memory from tropical cyclones can trigger heatwaves following flooding events ([Parker et al., 2013](#), [Chen et al., 2021](#)). Other phenomena, such as urban expansion and irrigation, may also exacerbate regional high temperature and precipitation conditions ([Wu et al., 2021c](#), [Mishra et al., 2020](#)).

However, the persistence of both types of events is also constrained by other factors. For instance, the duration of CHTDE is linked to the persistence of high-pressure systems; once cold air intrudes or a low-pressure system develops, this combination of high temperature and low precipitation will be broken ([Seneviratne et al., 2010](#)). In contrast, the continuation of CHTWE is typically associated with sustained moisture transport, active convective systems (such as monsoon rainfall, Meiyu fronts, or the peripheral circulations of typhoons), or prolonged frontal rainfall. These weather systems inherently possess longer lifecycles. For example, the continuous northward push of the East Asian Summer Monsoon and the extended activity of the Meiyu front can maintain warm, and heavy precipitation in a certain region for days or even weeks ([Tao, 1987](#)). Once these systems are established, their inertia can lead to their persistence for extended periods, provided that energy and moisture supply remain sufficient ([Zhou and Yu, 2005](#)). This might explain why CHTWE exhibit longer durations in this study.

4.3. Research limitations and future prospects

There are some considerations to keep in mind when interpreting the results of this study. In our research, the definition of CEs relies on variations in temperature and precipitation, and changes in their thresholds (Fig. 3), which directly influence the spatial distribution of the CE (Fig. 4, Fig. 5). Naturally, due to differences in the methods used to define CEs, results may vary slightly. For instance, differences in threshold selection (Wu et al., 2019b, Yang et al., 2023), variations in measurement indicators (Wu and Jiang, 2022, Anderson and Bell, 2011, Liao et al., 2018), and the choice of time scales (Wang et al., 2023) can all lead to discrepancies in outcomes. Additionally, this study evaluated CN05.1 data alongside CMIP6 multi-model simulations; previous research has confirmed that CMIP6 performed well in simulating CEs in northern hemisphere land areas (Ridder et al., 2021) and provides a robust assessment of extreme wet-hot event processes under multi-model ensembles (MME) (Wu et al., 2021a). However, uncertainties in model simulations may still lead to errors. For example, the study of Abdelmoaty et al. (2021) indicated that over 70% of CMIP6 models exhibit a maximum bias of approximately 10% in daily precipitation

The present study focused on the changing characteristics of two CEs, but we did not assess the underlying factors causing these changes. Currently, we primarily relied on maximum temperature and precipitation data to evaluate CEs. Incorporating additional indicators, such as VPD, soil moisture, wind speed, etc., could enhance the detail and reliability of the assessment. However, the simulation capabilities of these data will greatly limit explorations of future CEs, and the complexities of separating and weighting the interactions among different time scales and physical meteorological factors also pose challenges to assessment work. In future research, in addition to addressing these limitations, we will place greater emphasis on interdisciplinary intersections, such as exploring the impacts of different types of CEs on ecosystems, which may provide clearer insights into the damages caused by CEs. Such efforts will help us better understand the effects of climate warming on extreme climate events in China, providing a scientific basis for developing policies to address climate change and mitigate disaster risks.

5. Conclusion

This study assessed the occurrence characteristics and risks of CHTDE and CHTWE in China under historical and future scenarios based on daily maximum temperature and precipitation data, which can more objectively reflect changes in compound events (CEs) and their persistence on a daily time scale. Moreover, we evaluated the return period of CEs based on copulas to emphasize the risk of CEs occurring in the future. The study reveals that with increasing greenhouse gas emission scenarios, all indicators for CHTDE and CHTWE in China exhibit significant upward trends. Historically, CEs were “concentrated” in July and August; however, future CEs will occur more frequently across a wider range of months. A staggering 87% and 93% of the country’s regions will experience more frequent, larger, and more intense of CHTDE and CHTWE. More importantly, the three-dimensional coupled analysis emphasizes the complex interdependencies among the duration, frequency, and intensity of CEs under different climate scenarios. Under the same return period, all three indicators continuously rise in future scenarios, indicating that these CEs will become more common and severe. Our research explores the physical mechanisms of occurrence and persistence for both types of events and underscores that, with global warming, CHTWE will pose a greater threat. The findings provide crucial insights into understanding the evolution of future CEs.

7. Consent for publication

Not applicable.

Author contributions

JM and WS designed the research; ML wrote the manuscript with the support of TL; PG performed the data processing and picture making; FS performed language editors and references. All authors contributed to the interpretation of the results and the writing of the paper.

9. Plain Language Summary

Compound events (CEs) pose significant threats to society and everyday life. This study focuses on two extreme weather events affecting China: compound high temperature and dry events (CHTDE) and compound high temperature and wet events (CHTWE). We utilized climate models to look at how often these events occurred in the past and what might happen in the future. Unlike previous research, this study evaluated CEs based on daily-scale data, which avoids the coarse identification associated with monthly scale data. Furthermore, we emphasized the interdependencies among three key aspects of these CEs through a three-dimensional copula model. Our findings indicate that both types of events are expected to become more frequent, last longer, and be more intense across China, especially for CHTWE. While these events mainly occurred during the summer historically, they are projected to occur more frequently in a wider range of months in the future. We also found that the three-dimensional copula results more prominently highlight the severity of future CEs compared to the two-dimensional copula results, we need to pay close attention to the risks brought about by the CEs.

CRedit authorship contribution statement

Min Liu: Writing – review & editing, Writing – original draft, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Tao Li: Writing – review & editing, Validation, Resources. **Peter Goethals:** Writing – review & editing, Formal analysis. **Fengjiao Song:** Writing – review & editing, Methodology, Investigation. **Jianying Ma:** Validation, Funding acquisition, Conceptualization. **Wei Sun:** Funding acquisition, Conceptualization.

Ethics approval and consent to participate

Not applicable.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This research work was supported by the the Key Projects of Science and Technology Development Plan of Jilin Province ([20230303008SF](#)), the Scientific Research Project of Education Department of Jilin Province ([JJKH20250319BS](#)), the Natural Science Foundation of Jilin Province ([20230101171JC](#)), the National Natural Science Foundation of China ([31870456](#)).

Appendix A. Supplementary data

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The following are the Supplementary data to this article:

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Supplementary Data 1.

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Supplementary Data 2.

Data availability

Daily temperature and precipitation data from CMIP6 global climate models (GCMs) including CanESM5, FGOALS-g3, GFDL-CM4, IPSL-CM6A-LR, MPI-ESM1-2-HR, and MRI-ESM2-0, as well as Shared Socioeconomic Pathways (SSP) scenarios (SSP245 and SSP585), were obtained from the CMIP6 archive. The relevant download link can be found in [Zhou et al. \(2023\)](#) at <https://esgf-node.llnl.gov/search/cmip6/>. Additionally, daily maximum temperature and precipitation data from the CN05.1 dataset covering the period from

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